

Theory of Long-range Coherence in Biological Systems. I. The Anomalous Behaviour of Human Erythrocytes

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In this paper, an explicit expression of the interaction potential has been obtained, based on the Fröhlich model of long-range coherence in biological cells. These theoretical expressions are used to correlate with the experimental data obtained from the light scattering and the Brownian motion for the red blood cells (human erythrocytes). The necessary conditions are derived for the formation of rouleaux in human erythrocytes in the presence of long range interactions. The kinetics of rouleaux formation in terms of the properties of the diffusion coefficient has been investigated. It is concluded that both types of experimental data can be interpreted satisfactorily by this model.

1. Introduction

One of the remarkable features of the biological systems is the existence of an extremely high degree of order. Since any form of order in nature must eventually be the result of forces acting amongst the subsystems, a great deal of speculation regarding the origin of these forces has taken place. In a series of papers Fröhlich (1968*a,b*, 1970, 1973, 1977) has investigated the origin of these forces from a quantum mechanical standpoint. A brief description of his model will be given later in this paper. Recently some experimental evidence supporting Fröhlich's model has begun to emerge and we shall focus our attention on two different types of experiments for which the theoretical analysis will be carried out.

These two types of experiments are as follows: (1) Rowlands, Sewchand & Enns (1982) have conducted a series of experiments related to the Brownian motion of the human red blood cells (erythrocytes) and observed an increased rate of rouleaux formation which they correlated with the long range forces of Fröhlich. These forces may be either attractive or

repulsive depending upon specific experimental conditions. The erythrocytes tend to form clusters referred to as rouleaux. The important feature about this phenomenon of rouleaux formation, is that the rate of process cannot be explained on the basis of pure Brownian motion of the erythrocytes. It is necessary to postulate the existence of long range forces of Fröhlich.

(2) In an entirely different experiment, involving quasi-elastic light scattering from suspensions of erythrocytes, one of us has observed the appearance of anomalous diffusion coefficients in a similar condition to the rouleaux formation. It is not at all surprising that the process of rouleaux formation is itself diffusion controlled and may be the consequence of anomalous diffusion. The experimental details of the light scattering experiments will be published elsewhere.

For the purpose of the theoretical investigations developed hereafter, we will adopt the following conjectures: (a) there is evidence of the unusual behaviour of biological systems for two types of experiments; (b) both experimental observations can be explained on the basis of a long range force of Fröhlich operating between the erythrocytes.

The "specific conditions" required for either rouleaux formation or anomalous diffusion are the following: (a) a constant supply of energy is needed which can be provided from the metabolic energy of the cell; (b) the existence of a transmembrane potential is required to polarize the molecules of the system.

If either of these two conditions is not satisfied, then the usual behaviour of the erythrocytes vanishes. Consequently, the interparticle potential must be sensitive to the two factors mentioned above. The details of the theoretical investigations are as follows.

2. Theory

It is convenient to consider the theoretical aspects of this problem in three stages: (1) the formulation of the mathematical form of the long range potential; (2) the development of a theory that suitably incorporates the potential into the kinetics of rouleaux formation; and (3) the inclusion of a long range potential into the formula for the diffusion coefficient of the erythrocytes. This will provide important information about the anomalous diffusion properties.

We now subsequently discuss the three stages outlined above:

THE LONG RANGE POTENTIAL

Fröhlich (1972) showed that the coupled frequency $\omega_{\pm}$ of two giant dipolar molecules separated by a distance r and possessing frequencies ω_1

and ω_2 is given by

$$\omega_{\pm}^2 = \frac{1}{2}(\omega_1^2 + \omega_2^2) \pm \sqrt{\frac{1}{4}(\omega_1^2 - \omega_2^2) + \frac{\beta^4}{\epsilon_{\pm}^2}} \quad (1)$$

where

$$\beta^2 = \frac{\gamma^2 e^2 (Z_1 Z_2)^{1/2}}{M r^3} \quad (2)$$

and γ is a positive quantity of the order of unity which depends upon various angles, e denotes the electronic charge, M is the mass, Z_1 and Z_2 are the numbers of elastically bound particles in the two molecules and $\epsilon'_{\pm}$ represents the real part of the frequency dependent dielectric constant determined at $\omega = \omega_+$ and $\omega = \omega_-$, respectively, that is:

$$\epsilon'_{\pm} = \epsilon'(\omega_{\pm}). \quad (3)$$

If the two dipolar molecules are identical, then under conditions of frequency resonance ($\omega_1 = \omega_2$), an interaction potential $U(r)$ exists between them expressed by

$$U(r) = \frac{\hbar}{4\bar{\omega}} \frac{e^2 Z r}{M} \left[\frac{1}{\epsilon'(\omega_+)} - \frac{1}{\epsilon'(\omega_-)} \right] \frac{1}{r^3} \quad (4)$$

where $\hbar$ is Planck's constant, $Z = Z_1 = Z_2$, $\bar{\omega} = \omega_1 = \omega_2$ reflecting the identical nature of the two molecules. If $\epsilon'(\omega_+)$ turns out to equal $\epsilon'(\omega_-)$, then the interaction potential $U(r)$ of (4) vanishes and equation (1) is reduced to a London-Van der Waals type with a much shorter range since the potential is now proportional to r^{-6} . It, therefore, follows that the long range Fröhlich potential is dependent upon the frequency of the coupled oscillators through the real part of the dielectric constant. Since the phenomena of rouleaux formation and anomalous diffusion are manifestations of the potential $U(r)$, it is clear that the quantity $\epsilon'(\omega)$ plays a cardinal role here. In the second paper of this series (Paul, Chatterjee & Tuszyński, 1983) we have presented a detailed derivation of $\epsilon'(\omega)$. Here we will simply apply the final equation for it which is:

$$\epsilon'(\omega) = 1 + \frac{4\pi^2}{V k T} \sum_{5,6} \mu_{5,6}^2 (n_6 - n_5) \frac{(a_{5,6}^2 + b_{5,6}^2 - a_{5,6}\omega)}{(\omega - a_{5,6})^2 + b_{5,6}^2}. \quad (5)$$

The summation indices are expressed numerically, thus 5 and 6 should be regarded as indices not numbers. V is the volume of the suspension of erythrocytes and T is the absolute temperature. Also:

$$\mu_{5,6} = |\langle 5 | \mu | 6 \rangle| \quad (6)$$

where μ is a dipole moment operator and

$$b_{5,6}^2 = \frac{|\chi|^4}{\pi^2 \hbar^4 (n_6 - n_5)^2} \Gamma_{5,6}^2(\hbar^{-1} \omega_{5,6}) \quad (7)$$

here χ is a phonon-heat bath coupling constant, $\Gamma_{5,6}$ the line width determined under the resonance conditions and n is the occupation number of the phonon state i . Finally,

$$a_{5,6} = \hbar^{-1} \omega_{5,6} \quad (8)$$

where $\omega_{5,6}$ is the difference in energy between the excited (5) and ground (6) states or $\omega_{5,6} = \omega_5 - \omega_6$. The details will be given in our second paper.

With the dielectric constant given by equation (5) and the frequencies $\omega_{\pm}$ by equation (1), the Fröhlich potential can now be fully defined.

So far we have obtained the form of the interaction between two giant dipoles but the relationship with the erythrocytes has not been discussed yet. A typical erythrocyte is an extremely complex object. Most important, however, is the existence of a very large electrical field across the cell membrane. This field produces intense polarization of the cell membrane resulting in the generation of the giant dipoles as envisaged in Fröhlich's theory. Furthermore, the cell membranes are not stationary objects but display various modes of vibrations which provide us with the features required from a long-range potential. In our model, we assume that the giant dipoles are located at the centre of mass of each erythrocyte. This is, of course, an assumption since the dipoles are actually located on the membranes with different orientations. The exact orientations and numbers of the dipoles are not available to us at the present time forcing us to adopt the simplified model and allow experiment to decide the validity of the assumptions. It is worth pointing out, perhaps, that since our interest lies with the long-range forces between the erythrocytes, the exact cellular details may not be of any significance.

Having obtained the analytical form of the potential acting between the erythrocytes, let us now turn our attention to the nature (attractive or repulsive) of the forces. The frequency ω appearing in equation (5) is a positive quantity and from the discussion given in section 6 of our subsequent paper we see that $\omega_5 > \omega_6$ or that

$$\omega_5 - \omega_6 > 0 \quad (9)$$

which ensures the positive nature of $a_{5,6}$ defined by equation (8). The quantities $b_{5,6}^2$ and $\mu_{5,6}^2$ are positive definite by virtue of their definitions.

From equation (1) the frequency ω_- is lower than ω_+ , i.e. $\omega_+ > \omega_-$. Let us define a gap Δ as

$$\omega_+ = \omega_- + \Delta. \quad (10)$$

In order to compute the sign of the Fröhlich potential, it is necessary to calculate the relative magnitudes of the dielectric constants ϵ'_+ and ϵ'_- . These two quantities, when substituted into (4), determine the sign of $U(r)$. To evaluate the difference $\epsilon'_+ - \epsilon'_-$ we employ equation (5) and get:

$$\begin{aligned} \epsilon'(\omega_+) - \epsilon'(\omega_-) \\ = -\frac{4\pi^2}{V k T} \sum_{5,6} \mu_{5,6}^2 \left[\left(d_{5,6} + \frac{a_{5,6}}{B_{5,6} + C_{5,6}} \right) \right] (n_6 - n_5) \end{aligned} \quad (11)$$

where we have adopted the following notation

$$B_{5,6} = (\omega_- - a_{5,6})^2 + b_{5,6}^2 > 0. \quad (12)$$

The above is a positive quantity by virtue of the positive definiteness of the addands. We also used:

$$C_{5,6} = \Delta^2 + 2(\omega_- - a_{5,6})\Delta. \quad (13)$$

Unfortunately, no *a priori* statement about the sign of this term can be made. In a special case of a large enough gap Δ such that:

$$\Delta^2 > 2|(\omega_- - a_{5,6})|\Delta \quad (14)$$

the $C_{5,6}$ is positive:

$$C_{5,6} > 0. \quad (15)$$

We assume that the prevalent conditions allow the inequality (14) to hold thus assuring inequality (15).

Let us define

$$A_{5,6} \equiv a_{5,6}^2 + b_{5,6}^2 - a_{5,6}\omega_-. \quad (16)$$

From this definition and the inequality (15) it follows that

$$\frac{A_{5,6}}{B_{5,6} + C_{5,6}} < \frac{A_{5,6}}{B_{5,6}}. \quad (17)$$

A positive quantity $d_{5,6}$ has been introduced as

$$d_{5,6} = \frac{A_{5,6}}{B_{5,6}} - \frac{A_{5,6}}{B_{5,6} + C_{5,6}} > 0. \quad (18)$$

The right hand side of equation (11) now appears as a sum of terms, each of which contains two factors. One of these factors appearing in the

bracket is positive hence the overall sign is determined by the difference in population numbers ($n_6 - n_5$). Recalling the fact that the state 5 corresponds to higher energy than that of the state 6 (equation (9)) we deduce that two situations are possible.

(a) If the population inversion occurs ($n_5 > n_6$), then

$$\frac{1}{\epsilon'_+} - \frac{1}{\epsilon'} < 0 \quad (19)$$

and the overall sign of $U(r)$ becomes negative, and the erythrocytes experience an attractive force. An attractive force operating amongst the erythrocytes is, therefore, a signal of population inversion and holds under non-equilibrium conditions.

(b) Under equilibrium conditions ($n_5 < n_6$) the reverse of inequality (19) must hold true:

$$\frac{1}{\epsilon'_+} - \frac{1}{\epsilon'} > 0. \quad (20)$$

This inequality leads to a repulsive force amongst the erythrocytes.

In the above discussion we have shown that the population inversion is of great importance in determining the nature of $U(r)$. We must, now, seek for the possible underlying mechanism causing such an inversion. It is not hard to see that metabolism must play an essential role in supplying the required energy and we are thereby led to an understanding of the prerequisite (a) mentioned in Section 1. Moreover, the reasons for the prerequisite (b) can also be found from the equations we have already developed. If, in fact, the transmembrane field is switched off, then the force causing polarizations and the appearance of the dipole moment μ has been abolished and the Fröhlich potential vanishes.

Having gathered all necessary information about the potential $U(r)$ we proceed to discuss the next stage, i.e. the kinetics of the rouleaux formation.

THE KINETICS OF ROULEAUX FORMATION

In the absence of an interaction potential the process of rouleaux formation is solely controlled by the random Brownian motion of the erythrocytes. From a mathematical standpoint this problem can be formulated in much the same way as the kinetics of diffusion controlled chemical reactions (Rowlands *et al.*, 1982). In this approach one solves the diffusion equation and calculates the flux to a spherical surface. The boundary conditions required here correspond to a zero density of erythrocytes within the spherical surface and a bulk concentration at large distances. Such an

approach has been applied to the phenomenon of colloidal coagulation by Smoluchowski and well-documented literature on the subject is available (see for example Chandrasekhar, 1943). In the event that long range forces are operative between the erythrocytes, however, it is essential to go beyond the diffusion equation. An adequate treatment for diffusion controlled chemical reactions involving interacting particles has been developed by Debye (1942) and is well suited to the present problem. While it is not our intention to present all the details it seems worthwhile to introduce the basic mathematical steps as they may contribute to a better understanding of the physical principles involved. The analysis is most conveniently introduced in the following steps.

(a) Consider two volume elements dV_A and dV_B located by the vectors $\mathbf{r}_A$ and $\mathbf{r}_B$, respectively, as shown in Fig. 1.

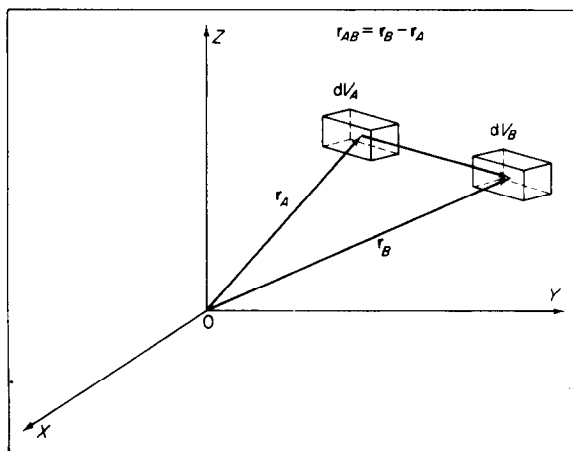


FIG. 1. Volume elements and coordinate systems.

(b) Let us introduce the following definitions. n_{AB} is the probability of finding an erythrocyte in dV_B given that dV_A already contains one erythrocyte. $N_A(\mathbf{r}_A)$ is the probability density of finding an erythrocyte in the vicinity of $\mathbf{r}_A$. N_{AB} is the joint probability of finding an erythrocyte in dV_A and dV_B .

By interchanging the labels A and B on n_{AB} and N_{AB} and by considering $N_B(\mathbf{r}_B)$ instead of $N_A(\mathbf{r}_A)$, one can interchange the physical role played by the two volume elements dV_A and dV_B .

From the above definitions it follows by straightforward physical arguments that the following equations must be true:

$$N_{AB} = N_A n_{AB} \quad (21)$$

$$N_{BA} = N_B n_{BA} \quad (22)$$

$$N_{AB} = N_{BA}. \quad (23)$$

(c) In completely general terms the functions introduced above must have the following spatial dependence:

$$N_{AB} = N_{AB}(\mathbf{r}_A, \mathbf{r}_B, \mathbf{r}_{AB}) \quad (24)$$

$$N_{BA} = N_{BA}(\mathbf{r}_B, \mathbf{r}_A, \mathbf{r}_{BA}) \quad (25)$$

and in the limit of infinite separation between the two volume elements:

$$\lim_{|\mathbf{r}_{AB}| \rightarrow \infty} N_{AB}(\mathbf{r}_A, \mathbf{r}_B, \mathbf{r}_{AB}) = N_A(\mathbf{r}_A) N_B(\mathbf{r}_B). \quad (26)$$

With no external gradients or fields present the probabilities defined above cannot depend upon the individual vectors $\mathbf{r}_A$ and $\mathbf{r}_B$ but upon the relative vector $\mathbf{r}_{AB}$ only:

$$N_{AB} = N_{AB}(\mathbf{r}_{AB}). \quad (27)$$

Consequently,

$$\lim_{|\mathbf{r}_{AB}| \rightarrow \infty} N_{AB}(\mathbf{r}_{AB}) = N_A N_B. \quad (28)$$

Clearly the erythrocytes are not stationary objects, hence in addition to the spatial dependence, each of the functions above must display time dependence as well.

(d) The joint probability function N_{AB} will, in general, satisfy a continuity equation:

$$-\frac{\partial}{\partial t} N_{AB} = \nabla_A \cdot \mathbf{J}_A + \nabla_B \cdot \mathbf{J}_B \quad (29)$$

where ∇_i are the gradient operators with respect to $\mathbf{r}_i$ and $\mathbf{r}_i$ are fluxes defined as

$$\mathbf{J}_i \equiv \mathbf{v}_i N_{AB} \quad (30)$$

$$\mathbf{v}_i \equiv \frac{d\mathbf{r}_i}{dt}. \quad (31)$$

(e) The fluxes must be selected in keeping with the physical requirements of the problem. In the present case the following two constituent fluxes are employed. First, a diffusional flux governed by Fick's law and involving

a diffusion coefficient D_i , and

$$\mathbf{J}_i^{\text{diff}} = -D_i \nabla_i N_{AB}. \quad (32)$$

Second, a flux caused by a force $\mathbf{K}_{AB}$ acting upon the erythrocyte located at $\mathbf{r}_A$, given that a second erythrocyte is located at $\mathbf{r}_B$. This flux is defined by the following set of equations:

$$\mathbf{J}_A^k = \mathbf{v}_A^k N_{AB} \quad (33)$$

$$\mathbf{v}_A^k = \frac{1}{\gamma_A} \mathbf{K}_{AB}. \quad (34)$$

Here, γ_A is the frictional constant of the erythrocyte at N_A .

Assuming that the forces $\mathbf{K}_{AB}$ and $\mathbf{K}_{BA}$ are due to a potential function U , such that

$$\mathbf{K}_{AB} = -\nabla_A U \quad (35)$$

$$\mathbf{K}_{BA} = -\nabla_B U \quad (36)$$

and inserting the total flux ($\mathbf{J}_i = \mathbf{J}_i^{\text{diff}} + \mathbf{J}_i^k$) along with the appropriate definitions, from equations (32)–(36) into equation (29), we obtain a modified diffusion equation:

$$\begin{aligned} \frac{\partial}{\partial t} N_{AB} = & D_A \nabla^2 N_{AB} + \frac{1}{\gamma_A} N_{AB} \nabla_A^2 U + \frac{1}{\gamma_A} (\nabla_A U)(\nabla_A N_{AB}) \\ & + D_B \nabla^2 N_{AB} + \frac{1}{\gamma_B} N_{AB} \nabla_B^2 U + \frac{1}{\gamma_B} (\nabla_B U)(\nabla_B N_{AB}). \end{aligned} \quad (37)$$

(f) In order to solve this equation with the aim of applying it to the present problem we first define a set of variables $\mathbf{R}$ and $\mathbf{r}$ as follows:

$$\mathbf{R} = \frac{D_A \mathbf{r}_A + D_B \mathbf{r}_B}{D_A + D_B} \quad (38)$$

$$\mathbf{r} = \mathbf{r}_A - \mathbf{r}_B. \quad (39)$$

Then, we make the following assumptions that U and N_{AB} do not depend upon $\mathbf{R}$ and that spherical symmetry prevails.

Under these conditions a very simple equation results from equation (36).

$$-\frac{\partial N_{AB}}{\partial t} = \nabla_r \cdot \mathbf{J}_{AB}^r \quad (40)$$

with

$$\mathbf{J}_{AB}^r = -(D_A + D_B) \frac{\partial}{\partial r} N_{AB} - \left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B} \right) \frac{\partial U}{\partial r} N_{AB}. \quad (41)$$

The J_{AB}^r represents the flux of erythrocytes along the vector $\mathbf{r}_{AB}$ where the origin of this vector has been placed at the origin of the coordinate system. By definition the flux is the current (I) per unit area. Thus, if the total surface area through which the erythrocytes move towards the origin is $4\pi r^2$, then

$$J_{AB}^r = \frac{I}{4\pi r^2}. \quad (42)$$

By equating the right hand sides of equation (41) and (42) we obtain

$$\frac{dN_{AB}}{dr} + \left(\frac{1}{D_A + D_B} \right) \left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B} \right) \frac{dU}{dr} N_{AB} = \frac{-I}{4\pi(D_A + D_B)r^2}. \quad (43)$$

If we assume that the erythrocytes "react" upon arrival at a critical spherical surface of radius r_0 , then equation (43) can be integrated to yield

$$N_{AB}(r) = -\frac{I}{4\pi(D_A + D_B)} \exp \left[-\left(\frac{1}{D_A + D_B} \right) \left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B} \right) U(r) \right] \\ \times \int_{r_0}^r dr' \frac{e^{(1/(D_A + D_B))} \left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B} \right) U(r')}{r'^2}. \quad (44)$$

(g) From equation (44) we require the current I . Not knowing the analytical form of $N_{AB}(r)$ we cannot do this computation in quite general terms. When $r \rightarrow \infty$ then equation (28) is true and if U vanishes asymptotically then we obtain:

$$N_A N_B = \frac{-I}{4\pi(D_A + D_B)} \cdot \frac{1}{r_0^*} \quad (45)$$

where

$$\frac{1}{r_0^*} \equiv \int_{r_0}^{\infty} dr \frac{\exp [(1/(D_A + D_B))(1/\gamma_A + 1/\gamma_B)U(r)]}{r^2}. \quad (46)$$

Solving equation (45) for I we have

$$-I = 4\pi(D_A + D_B)r_0^* N_A N_B \quad (47)$$

which is just the rate of "reaction" (rouleaux formation).

(h) Assuming that the erythrocytes A and B are identical we obtain the rate of erythrocyte depletion in the form of a simple second order kinetic equation

$$-\frac{dN}{dt} = 8\pi D r_0^* N^2 \quad (48)$$

which upon integration becomes

$$\frac{1}{N(t)} - \frac{1}{N(0)} = \frac{4kT}{3\eta} \left(\frac{r_0^*}{r} \right) t. \quad (49)$$

Here the Einstein definition of the diffusion coefficient along with the Stoke's expression for the friction coefficient have been adopted:

$$D = \frac{kT}{\gamma} \quad (50)$$

$$\gamma = 6\pi\eta r_0. \quad (51)$$

In the experimental work carried out by Rowlands *et al.* (1982), the kinetics of rouleaux formation has been studied by introducing an *ad hoc* parameter Z in the rate constant of a pure diffusion controlled process. This parameter is intended to account for the existence of an interparticle potential but the dependence of A upon the potential or even the form of the potential itself has not been explicitly stated. The expression used by these workers (Rowlands *et al.*, 1982) has the following form:

$$\frac{1}{N(t)} - \frac{1}{N(0)} = \frac{2kT}{3\eta} \Xi t \quad (52)$$

where $\Xi = Z/r_0$. Comparing equation (52) with equation (49) we notice that our parameter r_0^* plays precisely the same role as Z . The advantage, however, of using the generalized model employed by us lies in the fact that r_0^* is explicitly given in terms of Fröhlich's potential and the properties of this potential in the biological context have already been discussed.

(i) In order to link our theory with the experiments on rouleaux formation we must evaluate the integral in equation (46) which can be written as:

$$L(\alpha, r_0) = \int_{r_0}^{\infty} dr \frac{e^{-\alpha/r^3}}{r^2} \quad (53)$$

where

$$\alpha = \frac{1}{(D_A + D_B)} \left(\frac{1}{\gamma_A} + \frac{1}{\gamma_B} \right) A \quad (54)$$

and

$$U(r) = -\frac{A}{r^3}. \quad (55)$$

Since this integral cannot be solved analytically, we have to resort to the numerical integration instead. We treat α (or alternatively A) as an

unknown parameter we can vary and we calculate the corresponding values of the constant Ξ according to:

$$\Xi = \frac{1}{r_0 L(\alpha, r_0)} \quad (56)$$

We have used $r_0 \cong 4 \mu\text{m}$ as indicated by Rowlands *et al.* (1982). The results of our sample calculations for selected attractive potentials are presented in Table 1.

TABLE 1
Selected values of the parameter Ξ and the corresponding coefficients α and A .

α (10^{-9} cm^3)	A ($10^{-24} \text{ ergs cm}^3$)	Ξ
0.166×10^{-3}	0.007	1.11
0.011	0.444	1.16
0.021	0.848	1.21
0.029	1.172	1.25
0.422	17.049	2.66
0.521	21.048	2.91
0.770	31.108	3.45
0.914	36.926	3.73

It is not difficult to see that the integral of equation (53) gives the values of Ξ which are always less than 1 if the Fröhlich potential is repulsive ($A < 0$) and that Ξ tends to 1 as A approaches 0. On the other hand, it is the attractive potential ($A > 0$) that provides $\Xi \geq 1$. The Ξ increases with A starting at $\Xi = 1$ for $A = 0$. In their experiments, Rowlands *et al.* (1982) used a particle concentration of $5 \times 10^{-3} \text{ v/v}$ and the value of Ξ they arrived at was 3.0. As seen from Table 1, $\Xi = 3.0$ corresponds to $A \cong 23 \times 10^{-24} \text{ erg cm}^3$ or $\alpha \cong 0.57 \times 10^{-9} \text{ cm}^3$.

From our analysis employing a generalized diffusion equation and the Fröhlich potential we are able to explain the phenomenon of rouleaux formation and the dependence of the phenomenon upon the cell metabolism. Furthermore, by carrying out a numerical integration and using Rowlands' experiments data, the value of the constant in the Fröhlich potential has been obtained. Finally, the reduction of the rate of rouleaux formation to that expected under Brownian motion only by a lowering of the pH as observed by Rowlands *et al.* (1982) is explained by the destruction of the transmembrane field and the attending loss of polarization.

THE PHENOMENON OF ANOMALOUS DIFFUSION

Before we describe the experiments showing anomalous diffusion, some of the qualitative aspects of the diffusion coefficient in the presence of long range potentials must be introduced.

Pusey & Tough (1982) showed, that to a first order approximation, a non-stochastic interparticle potential $U(r)$ yields the following value of the mean square displacement (with the angular bracket denoting the average value):

$$\langle \Delta r^2(t) \rangle = 2D_0 t - \frac{D_0 t^2}{\gamma} \left\langle \frac{\partial^2 U(r)}{\partial r^2} \right\rangle \quad (56)$$

where D_0 is the diffusion coefficient of a Brownian particle in the absence of the potential. It is now possible to introduce a modified diffusion coefficient as follows

$$D = D_0 \left[1 - \frac{1}{2} \frac{t}{\gamma} \left\langle \frac{\partial^2 U(r)}{\partial r^2} \right\rangle \right]. \quad (57)$$

In applying this formula it is necessary to obtain the dependence of the potential $U(r)$ (in our case the Fröhlich potential) upon the particle density ρ . To achieve this objective let us examine equation (5). It is not hard to see that the value of summations must depend upon the total number of phonons N and therefore we write

$$\sum_{5,6} \mu_{5,6}^2 (n_6 - n_5) \frac{(a_{5,6}^2 + b_{5,6}^2 - a_{5,6}\omega)}{(\omega - a_{5,6})^2 + b_{5,6}^2} \cong NL \quad (58)$$

where L contains all the remaining quantities apart from N . With this approximation:

$$\varepsilon'(\omega) \cong 1 + \frac{4\pi^2}{kT} L\rho. \quad (59)$$

Consequently, the constant A in equation (55) to the first order approximation in the series expansion assumes the form

$$A = \frac{B}{\rho} + \dots \quad (60)$$

where B is a parameter containing a collection of constants. Thus, the modified diffusion coefficient becomes

$$D = D_0 \left[1 + \frac{6t}{\gamma} \left\langle \frac{B}{\rho r^5} \right\rangle \right]. \quad (61)$$

It is now clear that the attractive potential increases the value of D as compared to D_0 and *vice versa*.

In order to extract the density dependence of D we apply the first order of the virial expansion, i.e.:

$$\langle r^n \rangle \cong \rho^{-n/3} \quad (62)$$

and arrive at

$$\left\langle \frac{B}{\rho r^5} \right\rangle \cong B\rho^{2/3}. \quad (63)$$

We conclude that in keeping with our expectations, the coefficient D increases with the density ρ . This, in fact will account for the numerical differences between the Rowlands' results and the light scattering results.

To measure the diffusion coefficient of erythrocytes suspended in various solutions, the method of quasi-elastic light scattering has been employed. The 488 nm line of an Ar ion laser has been used and the scattered light has been detected through a pair of collimating pinholes by a photomultiplier. Then, the autocorrelation function plotted against time has given the diffusion coefficient as a slope.

The solution used is a phosphate buffered saline (PBS) at various pH to which human serum albumin (HSA) has been added. The additional macromolecules are polyvinylpyrrolidone (PVP) at several molecular weights. The PVP at the molecular weight of 360 000 causes an abnormally high diffusion coefficient when the blood substance is metabolically active and a transmembrane field is present. These are the conditions under which an increased rate of rouleaux formation was also observed by Rowlands *et al.* (1981). The only major difference between the two experiments is the particle concentration. In the light scattering experiments the concentration was 1.8×10^{-4} v/v, that is almost 28 times lower than that in Rowlands' experiments. Rowlands *et al.* (1981) obtained $\Xi = 3.0$ and in a comparable case, our result was:

$$D/D_0 = 1.56. \quad (64)$$

Table 2 shows all the results of measurements of the diffusion coefficient (in 10^{-10} cm²/sec) in various situations.

TABLE 2

The values of diffusion coefficient D in 10^{-10} cm²/sec for various solutions

Contents State	PBS		PBS + PVP		
	pH 7.4	pH 6.3	pH 7.4	pH 6.3	pH 7.4 [†]
Metabolically active	15.9	16.5	23.5	19.5	21.4
Metabolically depleted	15.4	14.8	17.5	19.0	18.1
Metabolically repleted	—	16.2	22.2	19.3	20.6

[†] Restoration of the pH of the suspension at 7.4 after a measurement had been taken at pH = 6.3.

In spite of the seeming discrepancy between the results of the measurements in two different types of experiments (Rowlands *et al.* (1981) $\Xi = 3.0$, light scattering $D/D_0 = 1.56$), these numbers are in good agreement when the density corrections are made. Another correction we have to make is the dimensionality factor. Rowlands *et al.*'s experiments (1981, 1982) treated the observed solution as a two-dimensional fluid whereas the light scattering method accounts for all three dimensions. Therefore, the diffusion constant used by Rowlands after Swift & Friedlander (1964) has to be multiplied by a factor of two:

$$K' = \frac{4\pi kT}{3\eta} \Xi. \quad (65)$$

To compare the light scattering diffusion constant with the corresponding result of Rowlands (1981) we therefore require that:

$$2 \cdot \Xi - 1 = \frac{D}{D_0} - 1. \quad (66)$$

However, when the density variation is accounted for we expect that:

$$\frac{2\Xi - 1}{D/D_0 - 1} = \left(\frac{\rho_R}{\rho_D} \right)^{2/3} \quad (67)$$

where ρ_R denotes the density in Rowlands' experiment (1981) and ρ_D is the density in the diffusion experiment. Substituting the relevant numbers we obtain a remarkable agreement as the left hand side produces 8.9 and the right hand side gives 9.2 which indicates the compatibility of the two methods.

3. Conclusions

In this paper we have discussed and extended some important aspects of the Fröhlich theory of the long range coherence in biological systems. We have adopted the assumptions of the Fröhlich model as our basis and using our derivations supplemented the missing expressions for the dielectric constants. The Fröhlich long range potential has, therefore, been explicitly defined mathematically.

We have discussed the necessary conditions for this potential to become attractive. All relevant algebraic expressions for the dielectric constants, frequencies and coupling coefficients have been given. We have found that the population inversion of the cell's energy levels is possible if only the energy supply is maintained above a certain threshold level and if the phonon-heat bath coupling constant is large enough.

We have given a brief review of the diffusion process for particles with appreciable mutual interactions. We have stated the formula for the rate constant in such processes. Combining this result with the analytical expression for the Fröhlich potential we have obtained information about the diffusion constant in rouleaux formation. Our numerical estimates of the rate constant for the rouleaux formation provided us with the values of the Fröhlich potential parameter and a threshold condition for the appearance of an increased rate of rouleaux formation.

We have carried out an analysis of the density dependence of our diffusion constant and concluded that to a first order approximation, D is proportional to $\rho^{2/3}$.

We have reported some experimental work done by one of us on the diffusion coefficient of human erythrocytes with the use of quasi-elastic light scattering technique. The experimental findings reveal a marked difference in the diffusion coefficient between the metabolically active and metabolically depleted cells. The process of metabolic depletion appears to be reversible. The presence of a transmembrane potential has proven to be another crucial factor in the cell behaviour which agrees with the theoretical predictions.

Finally, we have applied the theoretically derived expressions to interpret both types of experiments. We have compared the light scattering results with the corresponding results reported by Rowlands *et al.* (1982) and attempted to explain a seemingly paradoxical difference in the values of diffusion coefficient. When the differences in densities of the solutions and the dimensionality are taken into account, both results agree to within an astonishing 5%.

We conclude that the two experimental results can be interpreted by the same theoretical model. Furthermore, the proposed theory seems to provide a satisfactory explanation of the process of rouleaux formation in both cases.

We hope that applications of our model may well exceed the scope we have touched upon in this work.

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